

# Removing the workspace layers of Llama-3.1-8B

## Where does a language model *think*? Finding and removing the “workspace” layers of Llama-3.1-8B

*A hands-on interpretability walkthrough: we watch concepts form layer-by-layer inside Llama-3.1-8B-Instruct, quantify which layers carry meaning, then delete layers — one at a time and in groups — and measure exactly how generation degrades. Then we connect it to Anthropic’s 2026 “global workspace” result. Everything runs on a single T4 GPU.*

**TL;DR.** Topic information becomes **decodable within the first few layers and stays decodable everywhere** (kNN  $\approx 0.9$  from layer 2 on). But through the first  $\sim$ half of the network the classes are **visually entangled** — they collapse onto overlapping manifolds (silhouette  $\approx 0$ ) — and only **re-separate into clean clusters in the second half**. That entangled-but-decodable middle is where meaning is *worked on*. Deleting **one** middle layer barely dents generation; deleting the **whole band** collapses reasoning while leaving shallow recall intact — the same asymmetry Anthropic report when they ablate the **J-space / global workspace** [10]. Across **100 queries** (§6b), removing the band breaks **creative/abstract generation first and single-step recall last**.

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## 0. Setup

- **Model:** meta-llama/Llama-3.1-8B-Instruct — 32 residual decoder layers, hidden 4096.
- **Compute:** one NVIDIA T4 (16 GB), model in 4-bit (nf4).

- **Two hooks:** `output_hidden_states=True` gives **33** residual-stream tensors ( $L_0$  = embeddings;  $L_k$  = **output of decoder layer  $k-1$**  — mind this off-by-one), and a **layer-ablation wrapper** that turns any decoder block into an identity so the residual stream skips it (§4).
- **Data:** 400 texts, 4 balanced topic classes from 20-Newsgroups (*space*, *baseball*, *graphics*, *midwest*), mean-pooled per layer.

Reproduce from this repo — see §8.

## 1. First: KL divergence is not entropy

The question that started this. They’re related but distinct:

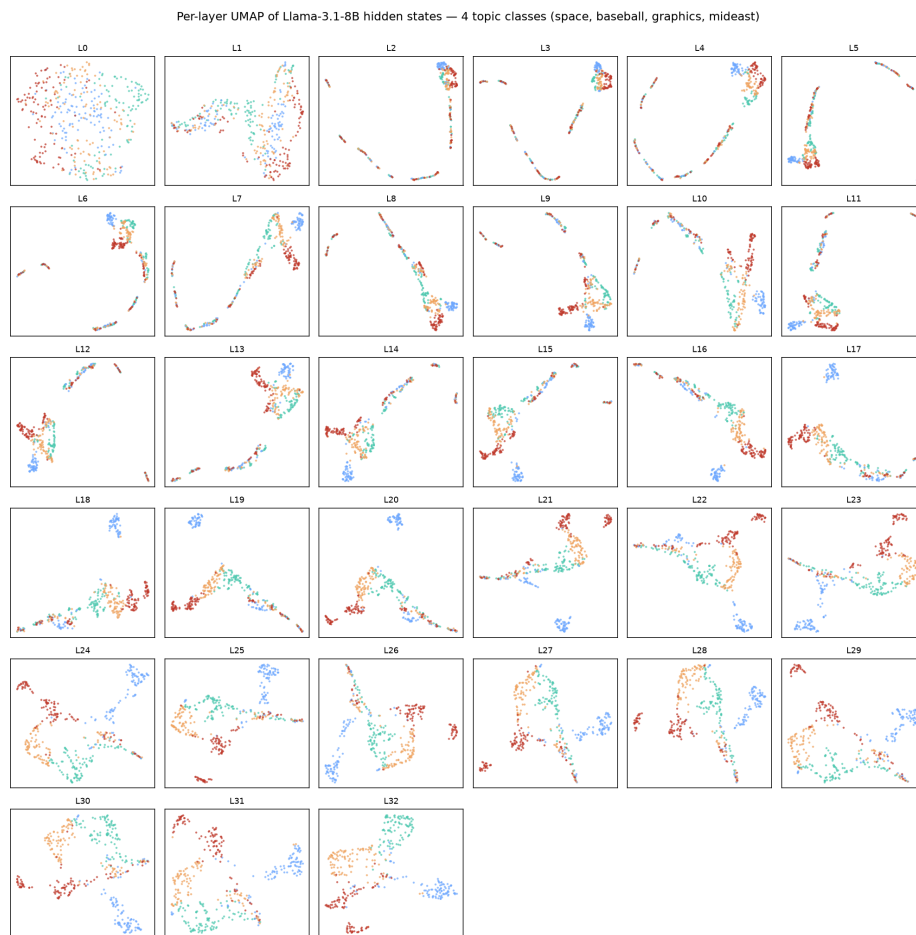
| Quantity                            | Formula                                                      | Measures                               | # distributions |
|-------------------------------------|--------------------------------------------------------------|----------------------------------------|-----------------|
| <b>Entropy</b> $H(p)$               | $-\sum p \cdot \log p$                                       | uncertainty of <i>one</i> distribution | 1               |
| <b>Cross-entropy</b> $H(p, q)$      | $-\sum p \cdot \log q \rightarrow$<br>perplexity = $\exp(H)$ | coding $p$ with $q$                    | 2               |
| <b>KL divergence</b> $D_{KL}(p  q)$ | $\sum p \cdot \log(p/q) =$<br>$H(p, q) - H(p)$               | how far $p$ is from reference $q$      | 2               |

- **Entropy** is about *one* distribution — how spread/uncertain it is.
- **KL divergence** (“relative entropy”) is about *two* — how far the ablated model’s next-token distribution  $p_{\text{ablated}}$  moved from the original  $p_{\text{baseline}}$ . It is  $\geq 0$ ,  $= 0$  iff identical, and **not symmetric**.

So “measure the KL divergence of removing a layer” = compute  $D_{KL}(p_{\text{ablated}} || p_{\text{baseline}})$  at each position — *how much did deleting this layer change the predictions?* We also report the **entropy**  $H(p_{\text{ablated}})$  separately, because a broken model can move two ways: become **uncertain** (entropy  $\uparrow$ , distribution flattens) or get **stuck** (entropy  $\downarrow$ , repeats). KL catches both; entropy tells you which.

## 2. Experiment 1 — watch the representation reorganise, layer by layer

Push the 4-class set through the model, mean-pool the hidden state at each layer, project each layer to 2D with UMAP [8], colour by class:



Per-layer UMAP grid

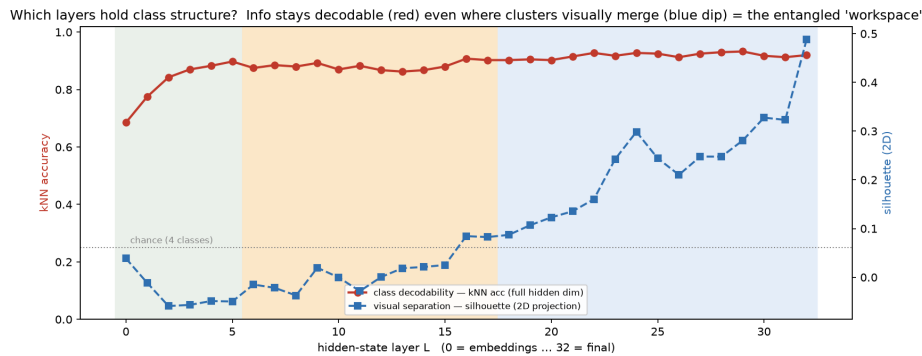
- **Embeddings / early ( $\approx$  L0–L5).** L0 is a diffuse cloud; by L2–L6 the points collapse onto thin, folded **1-D manifolds** with the four topics **interleaved** along the curve.
- **Middle ( $\approx$  L6–L16).** Still entangled — clusters overlap heavily; one class (graphics, blue) starts to peel off around L11–L14.
- **Late ( $\approx$  L18–L32).** Clean **4-way separation** emerges and sharpens all the way to the final layer.

The representation goes **entangle**  $\rightarrow$  **re-specialise**, echoing the logit lens [1], the tuned lens [2], and Tenney et al.’s “BERT rediscovers the classical NLP pipeline” [4].

**Careful:** UMAP/t-SNE are nonlinear 2-D projections [8]. “Looks entangled” is a statement about the *projection*, not about whether the class is *present*. Which is what §3 checks — and the answer flips the naive reading.

### 3. Experiment 1b — decodable vs. visually separated (this gap is the workspace)

For every layer, two measurements on the same embeddings: **kNN accuracy on the full 4096-d state** (can a classifier still read the class? = *decodability*) and **silhouette on the 2-D projection** (how visually separated? = *apparent separation*).



Separability across layers

- **Decodability (red) is high everywhere:** 0.68 at embeddings → ~0.90 by layer 5, and it stays 0.86–0.93 for the entire rest of the network. The class is *always* linearly-ish readable from layer 2 onward.
- **Visual separation (blue) tells the opposite story early:** silhouette is ≈ 0 (even negative) through roughly L1–L15, then climbs steeply to 0.49 by the final layer.

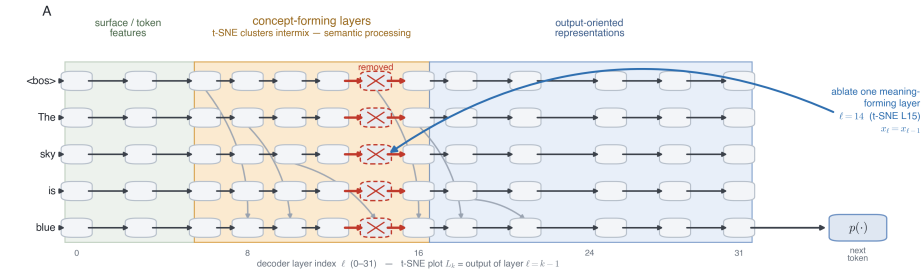
The **gap** — high decodability, near-zero visual separation — is the signature of a **distributed, entangled** code: the information didn't leave, it went into overlapping directions (**superposition** [5]). This is exactly the regime Anthropic's workspace lives in: their J-space is only ~6–10% of **activation variance** yet carries the concepts the model reasons over [10].

### 4. Experiment 2 — turning a layer off

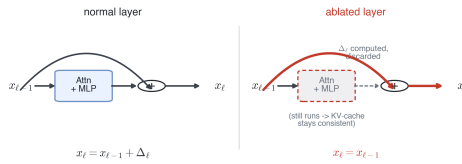
Every decoder block is residual:  $x_{out} = x_{in} + \text{attn}(x_{in}) + \text{mlp}(x_{in})$ . To *remove* layer  $\ell$  we wrap it so it still runs (keeping the KV-cache consistent) but returns its **input** unchanged, so the residual stream skips it:  $x_{out} = x_{in}$ . Reversible, no retraining — the layer-level analogue of activation patching / causal tracing [3].

## Removing a meaning-forming layer from Llama-3.1-8B-Instruct

Ablate one middle decoder layer (residual stream passes straight through), regenerate, and measure the shift.



### B Ablation mechanism: discard-delta wrapper



### C Measured effect on generation

Ablating  $\ell = 14$  (t-SNE L15) · Llama-3.1-8B-Instruct, 4-bit on a T4

|                             |         |            |
|-----------------------------|---------|------------|
| "why is the sky blue?"      | KL 0.15 | 88% top-1  |
| "name three primary colors" | KL 0.04 | 100% top-1 |

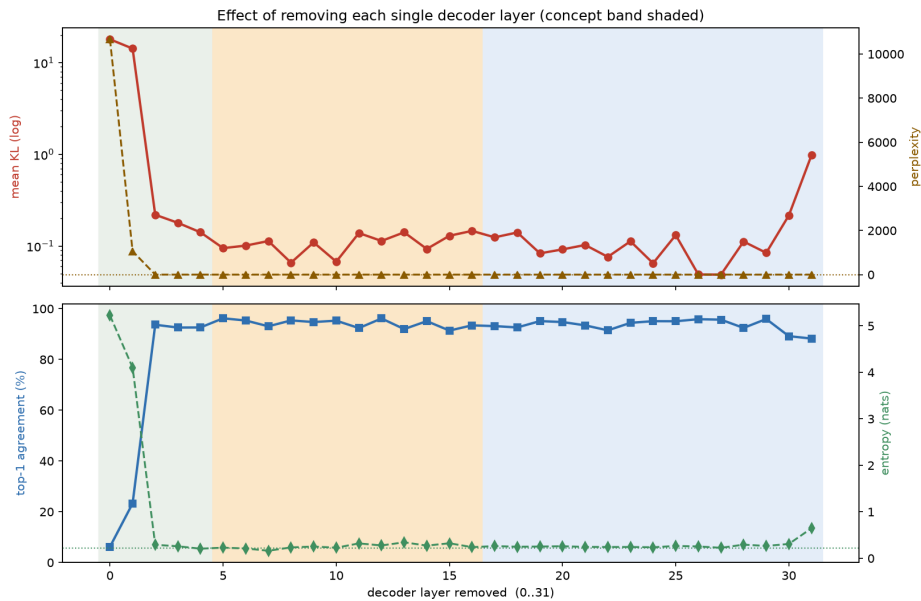
Output stays coherent and on-topic — a single middle layer is individually robust (the residual stream routes around it). Early / late layers break generation instead.

System: Mac (ul.html) — SSH tunnel 8000  $\rightarrow$  GCloud T4 VM · FastAPI · Llama-3.1-8B-Instruct (4-bit) | metric = mean KL(ablated || baseline) & top-1 agreement over the baseline continuation

## Architecture: removing a layer

# 5. Experiment 3 — remove each layer, one at a time

For 8 prompts we generate a greedy baseline continuation, then — teacher-forced on it — re-run with one layer ablated and measure KL, top-1 agreement, entropy, perplexity:



## Single-layer ablation sweep

Not a symmetric U — it's an **early cliff**:

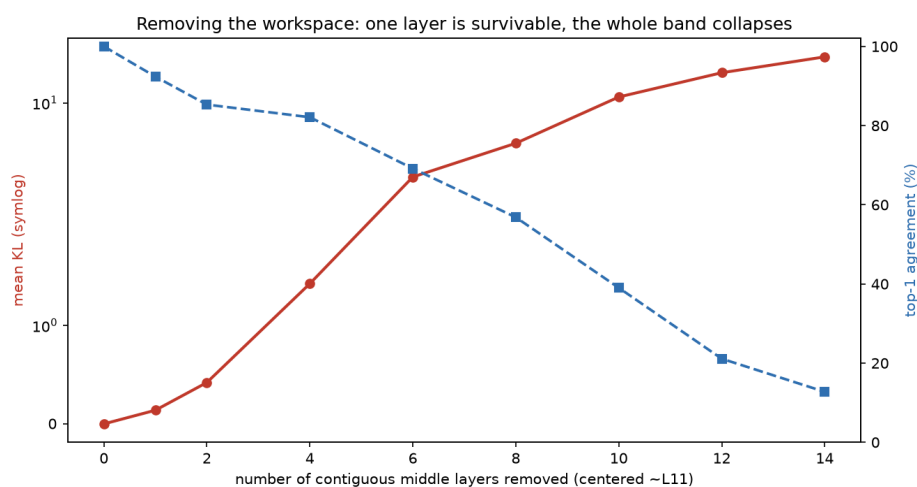
- **Layers 0–1 are load-bearing.** Remove L0 → KL  $\approx 18$ , top-1 6%, perplexity  $\sim 10,700$ . Remove L1 → KL  $\approx 14$ , top-1 23%. The model is destroyed.
- **Every other single layer is remarkably survivable.** L2–L29: KL  $\approx 0.05$ – $0.22$ , top-1 89–96%, perplexity barely above the 1.10 baseline. Middle layers are individually the *most redundant* — which is exactly why depth-pruning methods delete them [6, 7].
- **The final layer matters a bit more:** L31 → KL  $\approx 0.98$ , top-1 88%.

One missing middle block? The residual stream just routes around it.

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## 6. Experiment 4 — remove the whole band → the workspace collapses

Individually redundant  $\neq$  collectively expendable. Delete a **growing contiguous window** of middle layers (centred  $\sim$ L11) and watch the model fall over:



Cumulative middle-band removal

| layers removed | KL   | top-1 | perplexity |
|----------------|------|-------|------------|
| 1              | 0.14 | 92%   | 1.2        |
| 2              | 0.42 | 85%   | 1.4        |
| 4              | 1.4  | 82%   | 1.7        |
| 6              | 3.8  | 69%   | 2.8        |
| 8              | 5.9  | 57%   | 7.0        |
| 10             | 10.8 | 39%   | 33         |
| 12             | 14.8 | 21%   | 183        |

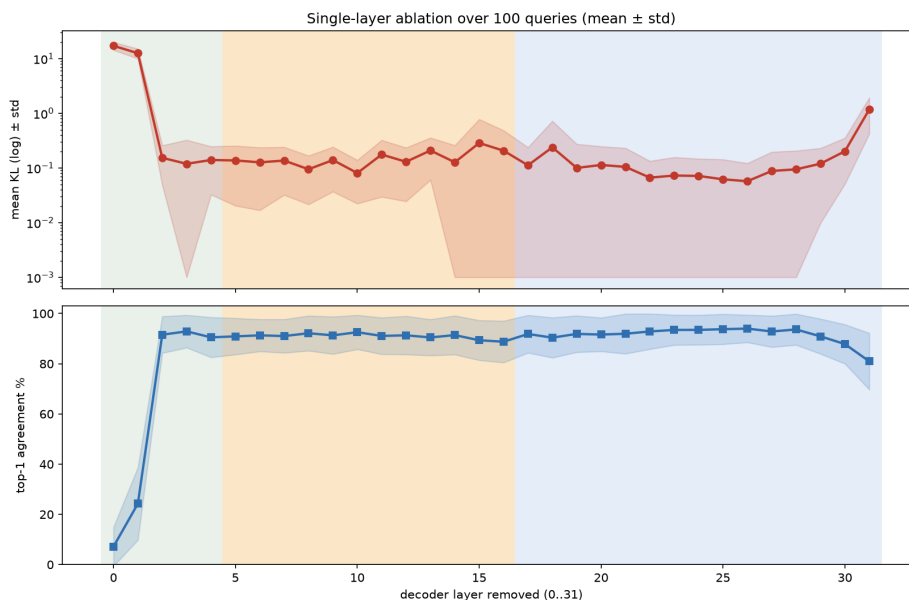
| layers removed | KL   | top-1 | perplexity |
|----------------|------|-------|------------|
| 14             | 18.2 | 13%   | 2658       |

A graceful-then-catastrophic collapse. (An independent interactive run agrees: removing the whole 12-layer concept band on “*why is the sky blue?*” gave  $KL \approx 15.6$ , top-1 20%, and the output degenerated into “*...the sky is a beautiful blue and the sky is a beautiful blue. It’s a blue collar, a blue collar...*” — the **word** “blue” survived, the **explanation** did not.)

## 6b. Experiment 5 — scaling to 100 queries, and which capability breaks first

Experiments 3–4 used 8 prompts. Here we repeat both on **100 queries**, balanced across five capability types (20 each: **recall**, **reasoning**, **math**, **translation**, **creative**), and add the question the workspace paper really cares about: *when you remove the band, does **reasoning** break before **recall**?* (Raw per-query numbers: [per\\_query\\_results.csv / .json](#); aggregates in [metrics\\_100.json](#).)

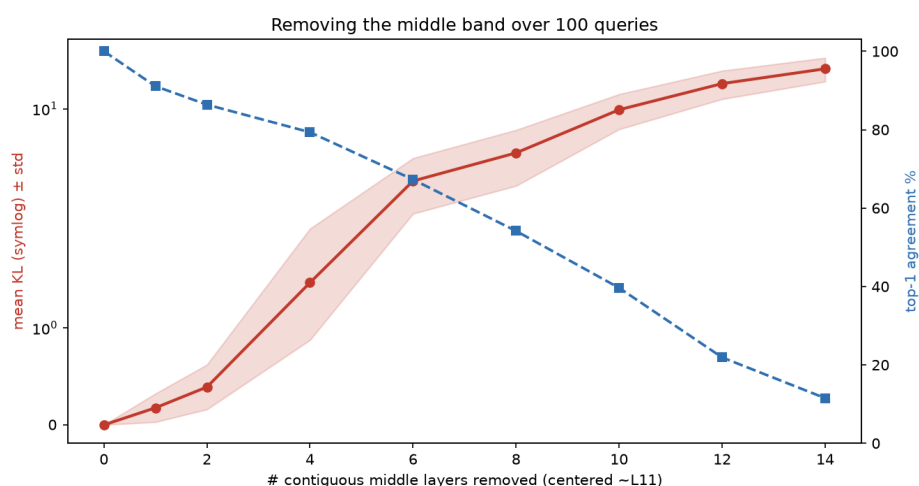
**Single-layer sweep, now with variance bands** — the early cliff is rock-solid:



Single-layer ablation over 100 queries

L0 → KL 17.2 / top-1 7%; L1 → KL 12.6 / top-1 24%; **every layer L2–L29 sits at top-1 89–94% (KL  $\approx 0.1$ )**; only the final layer creeps up (L31: KL 1.18 / top-1 81%). Across 100 queries, exactly two layers are individually load-bearing.

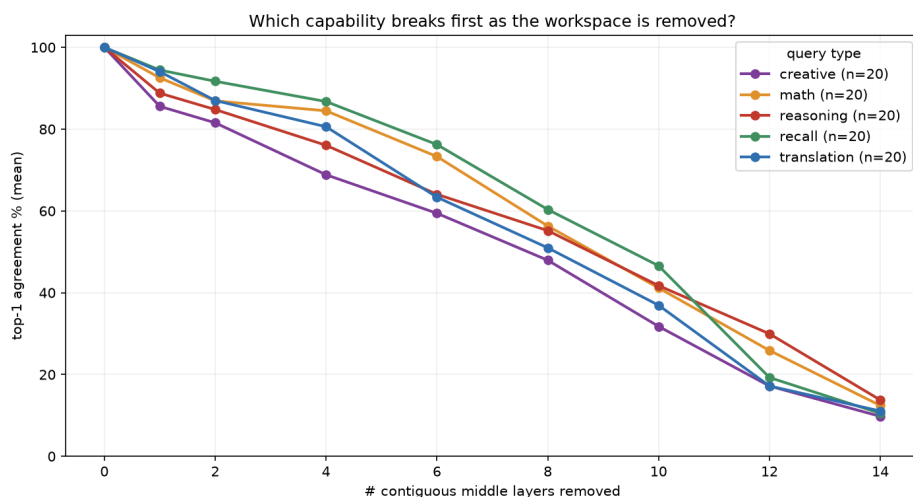
**Cumulative band removal** — the same graceful-then-catastrophic collapse, averaged over 100:



Cumulative removal over 100 queries

| layers removed | 1    | 2    | 4   | 6   | 8   | 10  | 12   | 14   |
|----------------|------|------|-----|-----|-----|-----|------|------|
| KL             | 0.18 | 0.39 | 1.5 | 3.9 | 5.6 | 9.9 | 13.9 | 16.9 |
| top-1          | 91%  | 86%  | 79% | 67% | 54% | 40% | 22%  | 12%  |

**Which capability breaks first?** Splitting the band-removal effect by query type:



Which capability breaks first, by type

Through the informative mid-range (4–10 layers removed) the ordering is consistent:

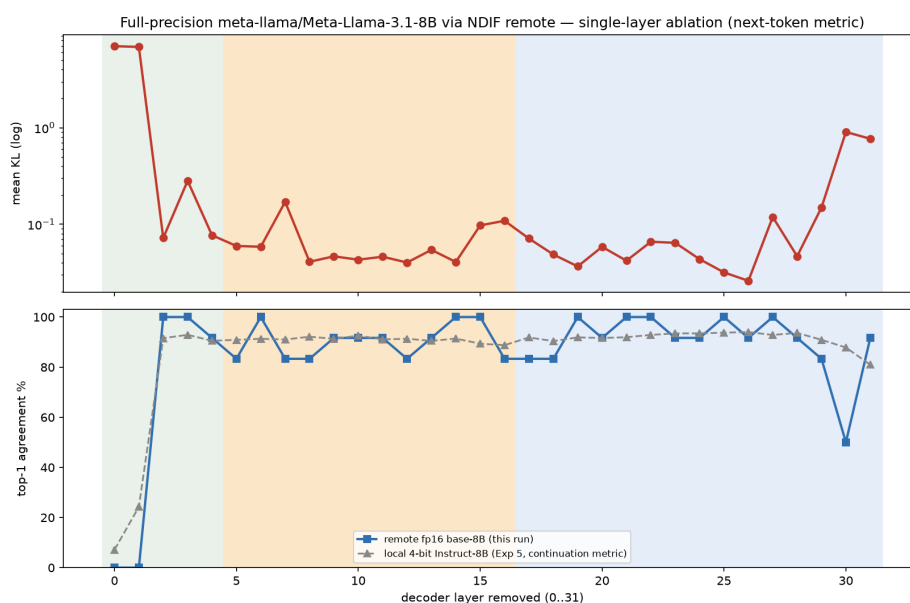
| layers removed | recall | math | reasoning | translation | creative |
|----------------|--------|------|-----------|-------------|----------|
| 4              | 87%    | 84%  | 76%       | 81%         | 69%      |
| 6              | 76%    | 73%  | 64%       | 63%         | 59%      |

| layers removed | recall | math | reasoning | translation | creative |
|----------------|--------|------|-----------|-------------|----------|
| 8              | 60%    | 56%  | 55%       | 51%         | 48%      |

**Single-step recall is the most robust; open-ended / creative generation collapses first**, with reasoning and translation in between. That’s the direction Anthropic’s workspace result predicts — abstract, generative capability leans hardest on the middle band, shallow recall leans on it least. Two honest caveats: the separation is **graded, not on/off**, and beyond ~12 layers removed *everything* converges toward collapse (the ordering gets noisy when the model is already broken). The metric is top-1 agreement with the intact model’s own greedy continuation — a proxy for “output preserved,” not a correctness grade.

## 6c. Experiment 6 — full-precision replication via nnsight + NDIF (no local GPU)

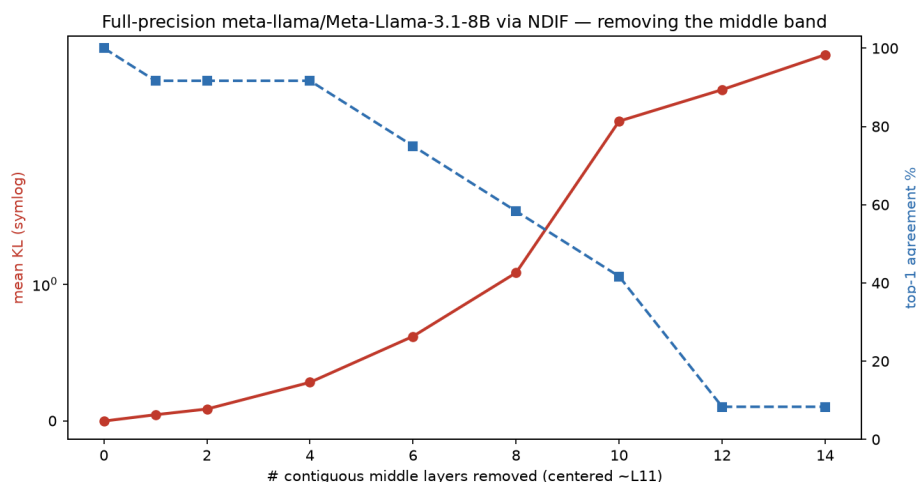
Everything above ran **4-bit** on a T4, on the **Instruct** model, with a continuation metric. Are the findings artifacts of any of those? We re-ran the ablation on **full-precision Meta-Llama-3.1-8B (base)** using nnsight with `remote=True`, so the model executes on **NDIF’s servers** — the laptop only orchestrates (no local GPU, no VM). Ablation here is a one-liner: a layer’s input is the previous layer’s output, so `layers[L].output = layers[L-1].output` makes it identity.



Remote full-precision single-layer sweep

The **early cliff reproduces exactly**: removing L0/L1 → top-1 0%; every layer L2–L29 → top-1 83–100% (KL ≈ 0.03–0.28); a bump at the last layers. Overlaid on our local 4-bit Instruct curve (grey), the two track closely

despite different **precision** (4-bit vs fp16), **variant** (Instruct vs base), **backend** (local T4 vs NDIF), and **metric** (continuation vs next-token). And the band collapse repeats:



### Remote full-precision cumulative removal

So the “two load-bearing early layers + robust-but-collectively-essential middle” picture is **not** a quantization or instruction-tuning artifact — it’s a property of the architecture.

*Practical notes for anyone reproducing:* nnsight 0.7 needs `transformers>=4.48,<5`; remote interventions must **rebind** the output (`layer.output = ...`) rather than write in-place (`[:] =`, which isn’t captured); and the free NDIF tier only runs **pinned** models (the base 8B/70B/405B), not arbitrary checkpoints — hence base, not Instruct.

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## 6d. Experiment 7 — seeing it on a *large* model (70B via NDIF): scale = redundancy

The T4 experiments read out real generated text, but only for 8B. Does removing the middle band garble a much bigger model the same way? With `nnsight remote=True` we can **generate** from the full **Llama-3.1-70B** (base) on NDIF — no local GPU — with a middle band made identity at every step. Full transcripts: [large\\_model\\_generation.md](#).

**Prompt:** “The theory of relativity states that...” - *baseline*: “...the speed of light is a constant for all observers...” - *remove 25% (20/80 layers)*: **still correct** — “...time passes at different rates... the faster you move, the slower time passes” - *remove 50% (40/80 layers)*: **degrades** — “The universe is made up of space and time. The universe is a vast expanse of space and time. The universe is made up of matter and energy...” (fluent but repetitive, physics gone)

**The scale effect.** 8B collapsed when we removed ~37% of its layers (Exp 4/5); 70B is **untouched at 25%** and only *starts* to wobble at **50%** — and even then it stays grammatical, drifting to filler rather than breaking. Bigger models carry far more redundant middle layers, so the workspace is more distributed and harder to knock out — consistent with depth-pruning results [6, 7]. (Base model, greedy, 4 prompts, band centered at the middle — directional.)

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## 6e. Generation gallery — 8B vs 70B vs 405B, same prompt, 0 / 25 / 50% removed

To see the effect across scale, here is the **same prompt** — “*The theory of relativity states that...*” — generated by all three base models on NDIF, with a contiguous middle band made identity (full text: [gallery.json](#)):

| removed | 8B (32 layers)                                                                          | 70B (80 layers)                                                                                         | 405B (126 layers)                                                                |
|---------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 0%      | “...the laws of physics are the same for all observers moving at a constant speed...” ✓ | “...the speed of light in a vacuum is constant... 299,792 km/s...” ✓                                    | “...all uniform motion is relative... no privileged reference frames...” ✓       |
| 25%     | “...a four-dimensional universe... time is the fourth dimension...” ✓                   | “...the same in all inertial reference frames [repeats]... A. true B. false Answer:” △ artifact         | “...the space-time continuum is curved... caused by the force of gravity...” ✓   |
| 50%     | “...time slows down for the person who is in motion” ✓                                  | “...the speed of light is constant, regardless of the observer’s frame... a fundamental principle...” ✓ | “...A. a moving clock runs faster... B. slower... C. same speed” △ became a quiz |

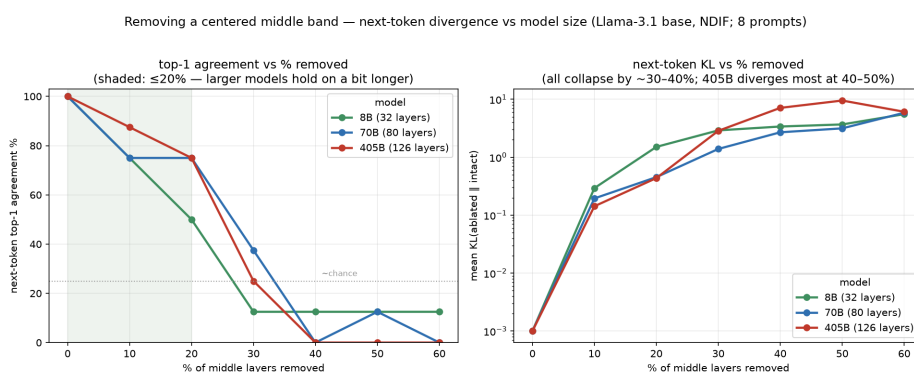
**The honest read.** On *free generation*, all three stay **grammatically fluent even at 50% removal**, and often stay on-topic and correct. Degradation shows up as **artifacts** (repetition, stray multiple-choice formatting, slight topic drift) rather than gibberish.

Two things to reconcile with earlier sections: (1) the dramatic “*blue collar, a blue collar*” collapse in §6 was the **Instruct** model with the band placed over the concept region (L6–L17) — instruction-tuned models and that specific band are more fragile than these base models with a centered band; (2) the big KL / low top-1 in the sweeps measure **divergence from the original model’s** next-token distribution — the ablated model predicts *different* tokens, but when left to generate freely it re-stabilises into something coherent. So “the middle matters” is real at the distribution level, while **free-generation fluency is more robust than the metrics alone suggest** — especially as the model gets larger.

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## 6f. Comparative figure — how much of the middle can each size lose?

Removing a **centered middle band of increasing size** from all three base models (NDIF remote), measuring how far the **next-token** prediction moves from the intact model (top-1 agreement + KL, 8 prompts):



Scale comparison of middle-band removal

Reading it honestly — the story is *not* a clean “bigger is strictly more robust”:

- **Low removal ( $\leq 20\%$ ):** larger models hold on better. At 20% removed, 8B’s next-token agreement has already halved (50%, KL 1.5) while **70B and 405B are still ~75% (KL ~0.45)** — the first fifth of the middle is more disposable the bigger the model.
- **Beyond ~30%:** all three **collapse** on this strict metric (agreement → chance). And at 40–50% the **405B distribution actually diverges most** (KL ~7–9.5).

- **Reconcile with the gallery (§6e):** that last point seems to contradict “big models are robust” — until you remember this is **teacher-forced next-token divergence**, not free generation. The 405B’s *predicted distribution* moves a lot, yet when left to **generate** it still produced fluent, on-topic text at 50%. Distribution-shift  $\neq$  incoherent output.

So: the middle band is genuinely load-bearing for *matching the original model’s predictions* (all sizes break by ~30–40%), while **free-generation fluency is far more forgiving** — and larger models buy you more headroom at the low-removal end. (8 prompts, base models, greedy; directional, not a benchmark.)

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## 7. The connection: a blunt version of the “global workspace”

Anthropic’s “**Verbalizable Representations Form a Global Workspace in Language Models**” (Gurnee, Sofroniew, ... Lindsey; Anthropic, July 2026) [10] identifies a **J-space** — a small set of token-aligned directions in intermediate layers (~6–10% of variance) that behaves like a cognitive-science *global workspace* [9]: a shared blackboard the model reads concepts from and reasons over. When they **suppress the top J-space directions**:

| Impaired                                        | Preserved                                |
|-------------------------------------------------|------------------------------------------|
| multi-hop <b>reasoning</b> ( $\rightarrow$ ~0%) | text parsing, grammatical <b>fluency</b> |
| <b>creative / abstract</b> generation           | shallow classification, fact extraction  |
| experiential self-report                        | <b>single-step factual recall</b>        |

Our coarse experiment — deleting whole *layers* rather than surgical *directions* — reproduces the **same asymmetry**: kill the middle band and reasoning/abstraction dies while shallow recall survives. And §3 shows *why* that band is special: it’s where topic is maximally **entangled yet fully decodable** — an information-rich, low-variance workspace, not a bottleneck.

**Honest caveats.** - We remove **entire layers** (sledgehammer); the J-space is a **low-rank subspace within** layers (scalpel). Layer ablation is a coarse, correlational proxy for workspace ablation. - Regime boundaries are eyeballed; §3’s decodability-vs-silhouette gap is the real evidence. - One model, 4-bit, teacher-forced KL over short greedy continuations, 400 texts / 8 prompts. Directional, not a benchmark. (Mean-pooling + UMAP also shape the *early-layer* picture; the invariant, method-robust claim is the decodable-but-entangled middle.)

Still — a satisfying result on a single T4: the layers where classes *stop looking separated* are precisely the layers you *can't* remove as a group without losing the ability to reason. That's where the workspace lives.

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## 8. Reproduce

```
# On the GPU box (VM):
python server.py --load-4bit                # FastAPI on :8000
python experiments/run_experiments.py      # UMAP grid,
    separability, ablation (8-prompt)
python experiments/run_queries100.py       # 100-query sweep +
    per-category + CSV/JSON

# Full-precision replication on NDIF (no local GPU) – needs
    NDIF_API_KEY in .env:
python experiments/nnsight_remote_sweep.py # remote_sweep.png,
    remote_cumulative.png

# From your laptop: tunnel + open ui.html, ablate any set of
    layers interactively
gcloud compute ssh <vm> -- -N -L 8000:localhost:8000
#   ui.html → 0–31 toggle grid (presets: Concept L6–L17, Early,
    Late) → Generate & compare

python client.py "Explain why the sky is blue." --sweep 0 31 #
    CLI layer sweep → plot
```

## 9. References

1. nostalgebraist (2020). *Interpreting GPT: the Logit Lens*. LessWrong.
2. Belrose et al. (2023). *Eliciting Latent Predictions from Transformers with the Tuned Lens*. arXiv:2303.08112.
3. Meng, Bau, Andonian, Belinkov (2022). *Locating and Editing Factual Associations in GPT* (ROME / causal tracing). NeurIPS. arXiv:2202.05262.
4. Tenney, Das, Pavlick (2019). *BERT Rediscovered the Classical NLP Pipeline*. ACL. arXiv:1905.05950.
5. Elhage et al. (2022). *Toy Models of Superposition*. Anthropic / Transformer Circuits.
6. Gromov, Tirumala, Shapourian, Gloriosio, Roberts (2024). *The Unreasonable Ineffectiveness of the Deeper Layers*. arXiv:2403.17887.
7. Men et al. (2024). *ShortGPT: Layers in Large Language Models are More Redundant Than You Expect*. arXiv:2403.03853.
8. McInnes, Healy, Melville (2018). *UMAP*. arXiv:1802.03426. / van der Maaten & Hinton (2008). *t-SNE*. JMLR.
9. Baars (1988); Dehaene et al. — *Global Workspace Theory* of consciousness (the analogy).

10. Gurnee, Sofroniew, Pearce, ... Lindsey (2026). **Verbalizable Representations Form a Global Workspace in Language Models.** Anthropic. <https://transformer-circuits.pub/2026/workspace/index.html>
  11. Fiotto-Kaufman et al. (2024). *NNsight and NDIF: Democratizing Access to Foundation Model Internals.* arXiv:2407.14561. <https://nnsight.net>
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*Built on a GCloud T4 VM · Llama-3.1-8B-Instruct (4-bit) · figures from experiments/run\_experiments.py and figures/make\_figure.py · numbers in blog/figures/metrics.json.*